# Category Theory and Functional Programming
# § 1.6 The Curry-Howard Correspondence

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# Preliminaries

## Convention

The number assigned to chapters, examples, exercises, figures, pages, sections, and theorems on these slides correspond to the numbers assigned in the textbook (Abramsky and Tzevelekos 2011).

# Outline

# Typed Lambda-Calculus

## Definition

A **typed $\lambda$-calculus** with function types, product types, unit type and empty type, is defined by:

(i) Types

$$
\begin{aligned}
\sigma, \tau ::= {} & b && \text{(base types)} \\
\mid {} & \sigma \to \tau && \text{(function types)} \\
\mid {} & \sigma \times \tau && \text{(product types)} \\
\mid {} & 1 && \text{(unit type)} \\
\mid {} & 0 && \text{(empty type)}
\end{aligned}
$$

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(ii) Terms

$$
\begin{array}{rcll}
M, N & ::= & x & \text{(variable)} \\
& \mid & \lambda x.M & (\lambda\text{-abstraction}) \\
& \mid & M\ N & \text{(application)} \\
& \mid & \langle M, N \rangle & \text{(pair)} \\
& \mid & \pi_1\ M \mid \pi_2\ M & \text{(projections)} \\
& \mid & \langle\rangle & \text{(unit)} \\
& \mid & \mathsf{absurd}\ M & \text{(absurd)}
\end{array}
$$

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(iv) Conversion rules

$$
\begin{aligned}
\lambda x.M &= \lambda y.M[\,x := y\,] && (\alpha\text{-conversion})\\
(\lambda x.M)\,N &= M[\,x := N\,] && (\beta\text{-conversion})\\
\lambda x.\,M\,x &= M,\ \text{with } x \text{ not in } M && (\eta\text{-conversion})\\
\pi_i\,\langle M_1, M_2\rangle &= M_i\\
M &= \langle \pi_1\,M, \pi_2\,M\rangle
\end{aligned}
$$

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# Typed Lambda-Calculus

(iii) Contexts

$$\begin{aligned} \Gamma ::= {} & \emptyset && \text{(empty context)} \\ \mid {} & \Gamma, x : \sigma, \ \text{with } x \text{ not in } \Gamma && \text{(context extension)} \end{aligned}$$

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(v) Typing rules

- Hypothesis

$$\frac{}{\Gamma, x : \sigma \vdash x : \sigma}\,\text{hyp} \qquad \text{(if } x : \sigma \in \Gamma\text{)}$$

- Function types

$$\frac{\Gamma, x : \sigma \vdash M : \tau}{\Gamma \vdash \lambda x.M : \sigma \to \tau}\,{\to}\mathrm{I} \qquad\qquad \frac{\Gamma \vdash M : \sigma \to \tau \qquad \Gamma \vdash N : \sigma}{\Gamma \vdash M\,N : \tau}\,{\to}\mathrm{E}$$

- Product types

$$\frac{\Gamma \vdash M : \sigma \qquad \Gamma \vdash N : \tau}{\Gamma \vdash \langle M, N \rangle : \sigma \times \tau}\,{\times}\mathrm{I} \qquad\qquad \frac{\Gamma \vdash M : \sigma_1 \times \sigma_2}{\Gamma \vdash \pi_i\, M : \sigma_i}\,{\times}\mathrm{E}$$

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# Typed Lambda-Calculus

(v) Typing rules

- Unit type

$$\frac{}{\Gamma \vdash \langle\rangle : 1}\ 1\mathrm{I}$$

- Empty type

$$\frac{\Gamma \vdash M : 0}{\Gamma \vdash \mathsf{absurd}\, M : \sigma}\ 0\mathrm{E}$$

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# Typed Lambda-Calculus

## Category associated to the typed $\lambda$-calculus

The category $\mathbf{TLC}$ is defined by:

(i) Objects: Types

(ii) Arrows: There is an arrow $\sigma \xrightarrow{M} \tau$ iff $M : \sigma \to \tau$.

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# Typed Lambda-Calculus

(iii) Composition

If $M : \rho \to \sigma$ and $N : \sigma \to \tau$ then

$$M \circ N : \rho \to \tau$$
$$M \circ N := \lambda x.\, M\,(N\,x)$$

(iv) Identity arrows

$$\mathrm{id}_\sigma : \sigma \to \sigma$$
$$\mathrm{id}_\sigma := \lambda x.x, \text{ with } x : \sigma$$

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(v) Unit axiom

Let $M : \sigma \to \tau$, then

$$\begin{aligned} M \circ \mathrm{id}_\sigma &= \lambda x. M(\mathrm{id}_\sigma\, x) \\ &= \lambda x.\, M\,((\lambda y.y)\, x) \\ &= \lambda x.\, M\, x \\ &= M \end{aligned} \qquad \begin{aligned} \mathrm{id}_\tau \circ M &= \lambda x.\, \mathrm{id}_\tau\, (M\, x) \\ &= \lambda x.\, (\lambda y.y)\, (M\, x) \\ &= \lambda x.\, M\, x \\ &= M \end{aligned}$$

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# Typed Lambda-Calculus

(vi) Associativity axiom

Let $M : \sigma_1 \to \sigma_2$, $N : \sigma_2 \to \sigma_3$ and $P : \sigma_3 \to \sigma_4$. Then

$$\begin{aligned}
M \circ (N \circ P) &: \sigma_1 \to \sigma_4 \\
M \circ (N \circ P) &= \lambda x.\, M\,((N \circ P)\,x) \\
&= \lambda x.\, M\,((\lambda y.\, N\,(P\,y))\,x) \\
&= \lambda x.\, M\,(N\,(P\,x)) \\
&= \lambda x.\, (\lambda y.\, M\,(N\,y))\,(P\,x) \\
&= \lambda x.\, (M \circ N)\,(P\,x) \\
&= (M \circ N) \circ P
\end{aligned}$$

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# Typed Lambda-Calculus

(vii) Initial object: Type $0$

$$\mathsf{i}_\sigma : 0 \to \sigma$$
$$\mathsf{i}_\sigma := \mathsf{absurd}$$

(viii) Terminal object: Type $1$

$$\mathsf{t}_\sigma : \sigma \to 1$$
$$\mathsf{t}_\sigma := \lambda x.\langle\rangle$$

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# Typed Lambda-Calculus

(ix) Binary products

We start with the objects and the arrows in the following diagram

$$
\begin{tikzcd}
\sigma_1 & \sigma_1 \times \sigma_2 \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & \sigma_2 \\
& \tau \arrow[ul, "M_1"] \arrow[u, dotted, "{(M_1, M_2)}"'] \arrow[ur, "M_2"'] &
\end{tikzcd}
$$

where $(M_1, M_2) := \lambda x. \langle M_1\, x, M_2\, x \rangle : \tau \to \sigma_1 \times \sigma_2$,

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# Typed Lambda-Calculus

First, we prove that the diagram is commutative.

$$
\begin{aligned}
\pi_i \circ (M_1, M_2) &: \tau \to \sigma_i \\
\pi_i \circ (M_1, M_2) &= \pi_i \circ \lambda y.\, \langle M_1\, y, M_2\, y \rangle \\
&= \lambda x.\, \pi_i\, ((\lambda y.\, \langle M_1\, y, M_2\, y \rangle)\, x) \\
&= \lambda x.\, \pi_i\, \langle M_1\, x, M_2\, x \rangle \\
&= \lambda x.\, M_i\, x \\
&= M_i
\end{aligned}
$$

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# Typed Lambda-Calculus

Next, we prove the uniqueness of arrow $(M_1, M_2)$. Let $N : \sigma_1 \times \sigma_2$ such that $\pi_i \circ N = M_i$. Then

$$
\begin{aligned}
(M_1, M_2) &= \lambda x.\, \langle M_1\, x, M_2\, x\rangle \\
&= \lambda x.\, \langle (\pi_1 \circ N)\, x, (\pi_2 \circ N)\, x\rangle \\
&= \lambda x.\, \langle (\lambda y.\, \pi_1\, (N\, y))\, x, (\lambda y.\, \pi_2\, (N\, y))\, x\rangle \\
&= \lambda x.\, \langle \pi_1\, (N\, x), \pi_2\, (N\, x)\rangle \\
&= \lambda x.\, N\, x \\
&= N
\end{aligned}
$$

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# Typed Lambda-Calculus

(x) Finitary products

Since $\mathbf{TLC}$ has binary products and a terminal object (nullary product) then it has finite products.

# References

# References

S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Bob Coecke. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: 10.1007/978-3-642-12821-9_1 (cit. on p. 2).